

CYBERSECURITY INTERNSHIP

Task 3 Lab Report

Web Application Security

Network Reconnaissance · Port Scanning · Packet Analysis · Firewall Configuration

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Organisation	ApexPlanet Software Pvt. Ltd.
Task	Task 3 – Web Application Security
Lab Environment	Kali Linux + Metasploitable2 (VirtualBox Host-Only Network)
Target IP	192.168.56.101 Attacker IP: 192.168.56.102
Date	June 2026
Tools Used	Nmap 7.98 · Wireshark 4.6.4 · iptables · ping · nslookup

■ **Lab Ethics Statement:** All activities documented in this report were conducted exclusively within an isolated VirtualBox virtual network. No external systems were targeted. This report is prepared for educational purposes as part of the ApexPlanet Cybersecurity Internship Programme.

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1. Executive Summary

This laboratory report documents the hands-on network security exercises conducted as part of Task 3 of the ApexPlanet Cybersecurity Internship. The objective was to apply core penetration-testing and network-defence techniques against an intentionally vulnerable target (Metasploitable2) running within an isolated VirtualBox environment.

The lab covered four interconnected phases: **(1) Host Discovery & Reconnaissance** using ping and nslookup; **(2) Port & Service Scanning** with multiple Nmap scan types to enumerate 23 open services; **(3) Packet Capture & Traffic Analysis** using Wireshark to inspect ICMP, ICMPv6, and IGMPv3 frames; and **(4) Host Firewall Configuration** via iptables to demonstrate traffic-blocking controls.

The target system (192.168.56.101) was found to expose **23 open TCP ports** running critically vulnerable software including a vsftpd 2.3.4 backdoor, an UnrealIRCd backdoor, Samba (EternalBlue-vulnerable), unauthenticated r-services, and an exposed Metasploitable root bind-shell. The OS fingerprint confirmed Linux kernel 2.6.9–2.6.33. Firewall rules were successfully applied to demonstrate HTTP port blocking.

2. Lab Environment & Topology

Component	Details
Host OS	Windows / macOS (VirtualBox 7.x host)
Attacker VM	Kali Linux 2024 – IP: 192.168.56.102
Target VM	Metasploitable2 (Ubuntu 8.04) – IP: 192.168.56.101
Network Type	Host-Only Adapter (vboxnet0) – no internet access
Nmap Version	7.98
Wireshark Version	4.6.4
Interface Used	eth1 (Kali → Metasploitable link)

3. Phase 1 – Host Discovery & Reconnaissance

Objective: Confirm that the target host is alive and reachable before beginning active port scanning.

Tools: ping (ICMP echo) · nslookup (DNS reverse lookup)

A ping sweep was launched against 192.168.56.101. The target responded to all 8 packets with 0% packet loss and average RTT of 2.091 ms, confirming the host was live. An nslookup reverse-DNS query returned NXDOMAIN, which is expected in an isolated lab (no DNS server configured). The ping response validates layer-3 connectivity between the attacker and target VMs before scanning commences.



Fig 1 – ping (8 packets, 0% loss, avg 2.091 ms) and nslookup NXDOMAIN on 192.168.56.101

4. Phase 2 – Port & Service Scanning

Multiple Nmap scan types were used progressively to enumerate open ports, identify running services, and determine exact version strings. Three scan modes were employed:

Scan Type	Command	Purpose
TCP SYN Scan	<code>nmap -sS 192.168.56.101</code>	Fast stealth scan – identifies open ports
Version Detection	<code>nmap -sV 192.168.56.101</code>	Probes services for version banners
OS Detection	<code>sudo nmap -O 192.168.56.101</code>	Fingerprints OS via TCP/IP stack analysis

4.1 TCP SYN Scan (nmap -sS)

The SYN scan completed in 0.95 seconds, discovering **23 open TCP ports**. Because SYN packets do not complete the full three-way handshake, this scan is considered semi-stealthy and is the default scan type for privileged users.

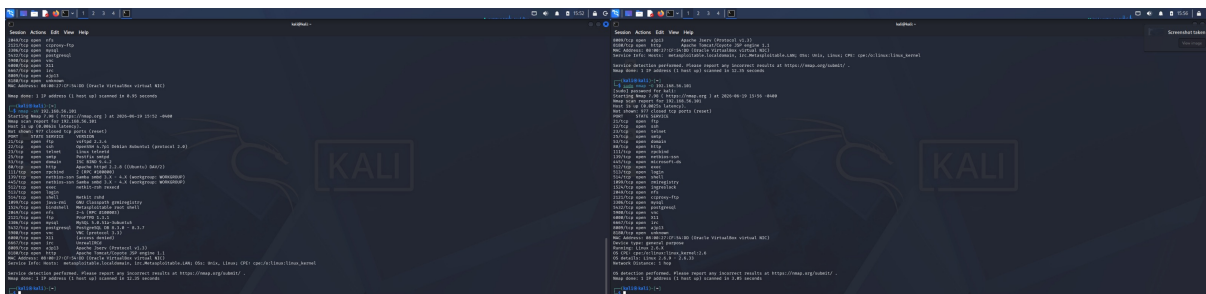


Fig 2 – nmap -sS: TCP SYN scan, 23 open ports, 0.95 sec

Fig 3 – nmap -sS confirmation run – identical 23-port result

4.2 Service Version Detection (nmap -sV)

The version scan probed each open port and retrieved exact service banners. Key findings include vsftpd 2.3.4 (known backdoor), OpenSSH 4.7p1 (deprecated), Apache httpd 2.2.8, Samba 3.x–4.x, MySQL 5.0.51a, PostgreSQL 8.3.0, VNC 3.3, UnrealIRCd (backdoor), and a Metasploitable root bind-shell on port 1524.

```

kali@kali:~$ nmap -sV 192.168.56.101
Starting Nmap 7.98 ( https://nmap.org ) at 2020-06-19 15:52 -0400
Nmap scan report for 192.168.56.101
Host is up (4.00ms latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
2117/tcp  open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
143/tcp   open  netbios-ssn
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  java-rmi
1524/tcp  open  bindshell
2049/tcp  open  nfs
2121/tcp  open  ftp
3386/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  x11
6667/tcp  open  irc
8009/tcp  open  ajp13
8188/tcp  open  http
MAC Address: 08:00:27:CF:54:D0 (Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.95 seconds

```

Fig 4 – nmap -sV: version scan showing vsftpd 2.3.4, OpenSSH 4.7p1, Apache 2.2.8, Samba, MySQL

Fig 5 – nmap -sV full listing: bindshell, NFS, ProFTPD, PostgreSQL, VNC, IRC

Fig 6 – nmap -sV second run: consistent results confirm reliable enumeration

```

kali@kali:~$ nmap -sV 192.168.56.101
Starting Nmap 7.98 ( https://nmap.org ) at 2020-06-19 15:52 -0400
Nmap scan report for 192.168.56.101
Host is up (4.00ms latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
2117/tcp  open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
143/tcp   open  netbios-ssn
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  java-rmi
1524/tcp  open  bindshell
2049/tcp  open  nfs
2121/tcp  open  ftp
3386/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  x11
6667/tcp  open  irc
8009/tcp  open  ajp13
8188/tcp  open  http
MAC Address: 08:00:27:CF:54:D0 (Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 13.41 seconds

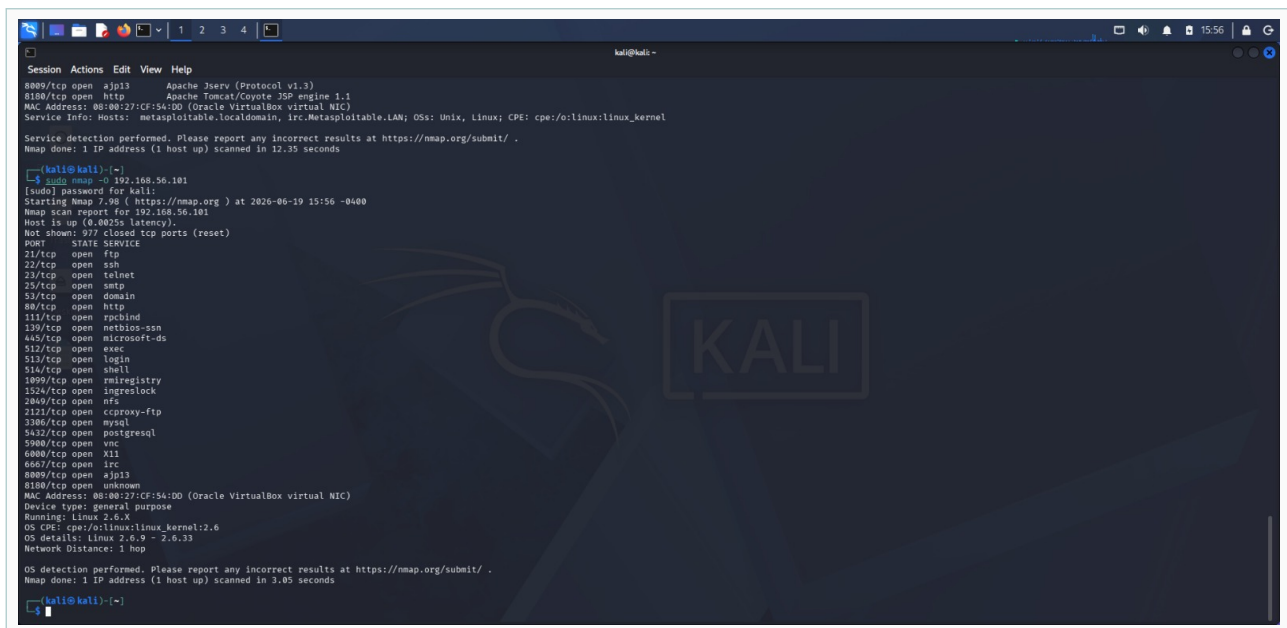
```

Fig 7 – nmap -sV lower panel: Samba, MySQL 5.0.51a, PostgreSQL 8.3.0, VNC 3.3, UnrealIRCd, Tomcat

5. Phase 3 – OS Fingerprinting (nmap -O)

OS detection uses TCP/IP stack behaviour analysis to infer the target's operating system. Nmap compared probe responses against its OS fingerprint database.

Result: OS fingerprinted as **Linux 2.6.9 – 2.6.33** (general-purpose device). CPE: *cpe:/o:linux:linux_kernel:2.6*. Network distance: 1 hop. This confirms Metasploitable2 is running a legacy Ubuntu kernel — consistent with the service banners retrieved (Ubuntu 8.04 / 8ubuntu1 strings).



```
Session Actions Edit View Help
kali@kali: ~
8000/tcp open  ajp13      Apache Jserv (Protocol v1.3)
8100/tcp open  http       Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:CF:54:00 (Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.35 seconds

kali@kali: ~
$ sudo nmap -h 192.168.56.101
[sudo] password for kali:
Starting Nmap 7.98 ( https://nmap.org ) at 2020-06-19 15:56 -0400
Nmap scan report for 192.168.56.101
Host is up (0.0025s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  cproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8100/tcp  open  unknown
MAC Address: 08:00:27:CF:54:00 (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 3.95 seconds

kali@kali: ~
$
```

Fig 8 – nmap -O: OS detected as Linux 2.6.X, CPE *cpe:/o:linux:linux_kernel:2.6*, 1-hop distance

6. Phase 4 – Packet Capture & Analysis (Wireshark)

Wireshark 4.6.4 was used to capture and inspect live network traffic on interface eth1. The capture workflow consisted of three steps: launching Wireshark, starting a live capture, and applying display filters to isolate specific protocol traffic.

6.1 Wireshark Launch & Interface Selection

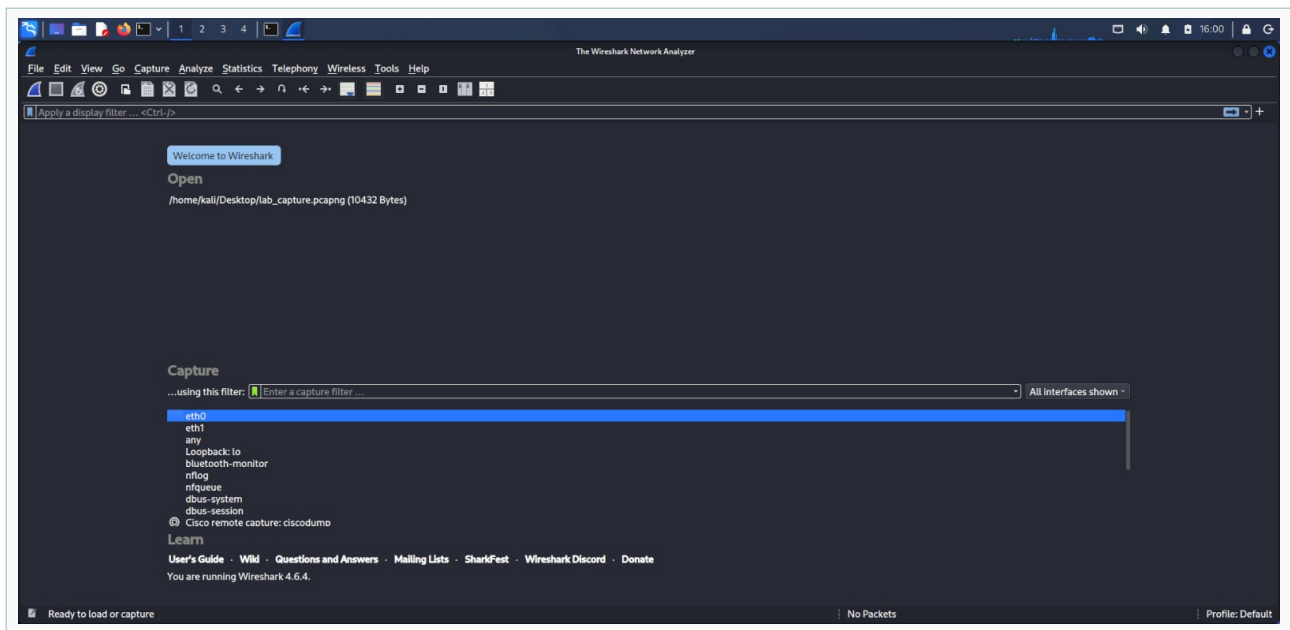


Fig 9 – Wireshark 4.6.4 start screen; eth1 selected as capture interface (host-only link)

6.2 Live Packet Capture

Capture was started on eth1. Initial frames included ICMPv6 Multicast Listener Reports (MLDv2) and IGMPv3 Membership Reports generated automatically by the Kali network stack. LLNMR (Link-Local Multicast Name Resolution) queries for *TUMUEXPERTBOOK* were also observed — these are Windows-style name resolution broadcasts from the host machine.

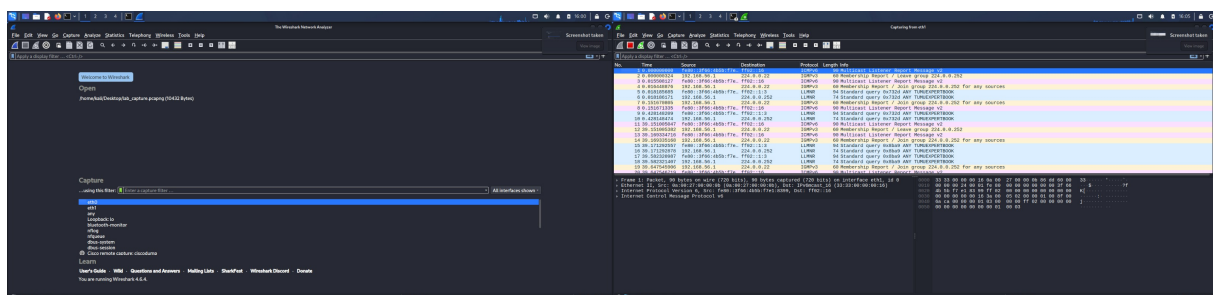


Fig 10 – Live capture: 72 packets; ICMPv6 MLDv2 and IGMPv3 visible

Fig 11 – Frame 1 detail: ICMPv6 Multicast Listener Report Message v2

6.3 ICMP Display Filter

A display filter **icmp** was applied to isolate ICMP echo (ping) traffic generated during the host-discovery phase. Wireshark showed request/reply pairs between 192.168.56.102 (Kali) and 192.168.56.101 (Metasploitable), with incrementing sequence numbers and consistent 98-byte frames — confirming the successful ping sessions logged in the reconnaissance phase.

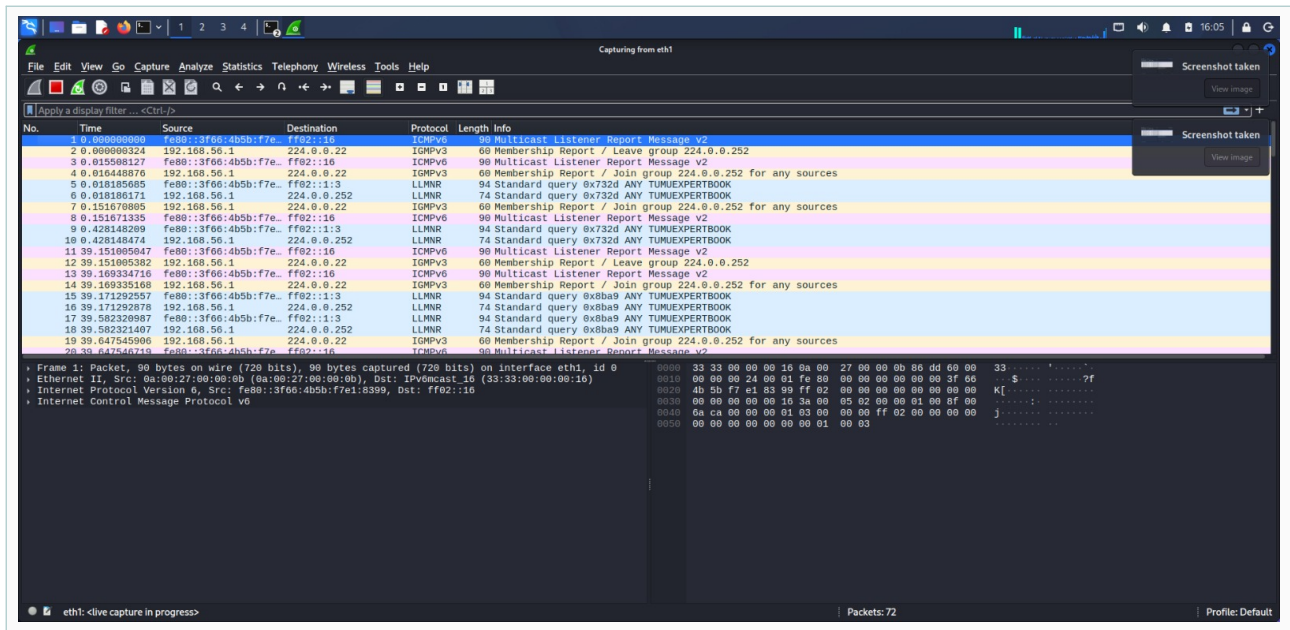


Fig 12 – Display filter 'icmp': echo request/reply pairs, seq 1–9, 98 bytes, TTL=64

7. Phase 5 – Firewall Configuration (iptables)

iptables is the Linux kernel's built-in packet-filtering firewall. This phase demonstrated how to inspect the current ruleset, add a traffic-blocking rule, and verify it was applied correctly — a core skill in host-based security hardening.

7.1 Baseline Ruleset (Empty Chains)

Before adding any rules, the existing iptables configuration was checked. All three chains (INPUT, FORWARD, OUTPUT) had policy ACCEPT with no rules — a completely permissive default state typical of a freshly booted Kali instance.

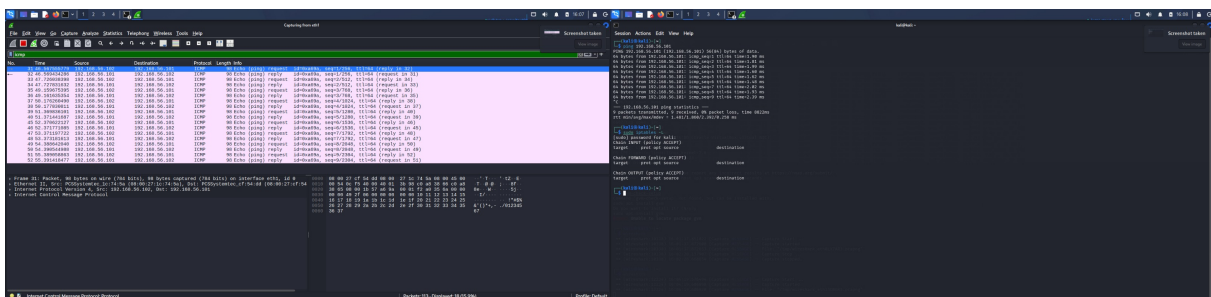


Fig 13 – `sudo iptables -L`: baseline — all chains empty, policy ACCEPT

Fig 14 – Second iptables -L check: clean state confirmed before rule add

7.2 Adding a DROP Rule for HTTP (Port 80)

Command used: `sudo iptables -A INPUT -p tcp --dport 80 -j DROP`

This appends a rule to the INPUT chain: any inbound TCP packet destined for port 80 (HTTP) is silently dropped. This simulates blocking web traffic to the Kali host.

After adding the rule, iptables -L was run again to verify the change. The INPUT chain now shows one rule: **DROP tcp -- anywhere anywhere tcp dpt:http**. The FORWARD and OUTPUT chains remain empty.

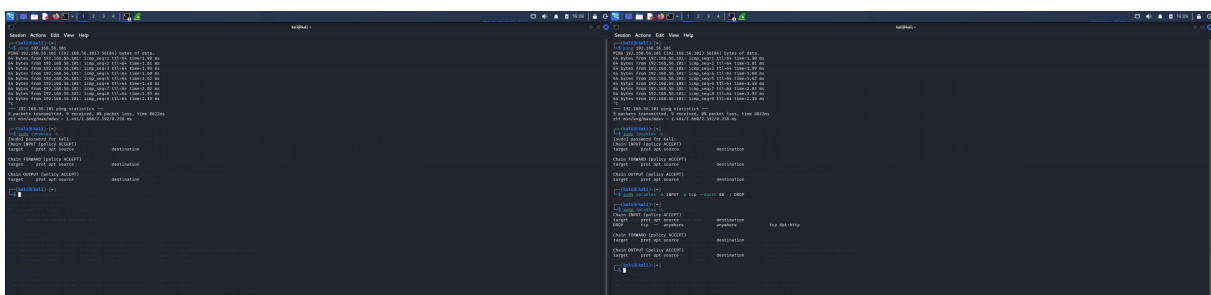


Fig 15 – Rule added and verified: DROP tcp dpt:http in INPUT chain

Fig 16 – iptables -L confirmation: HTTP drop rule active

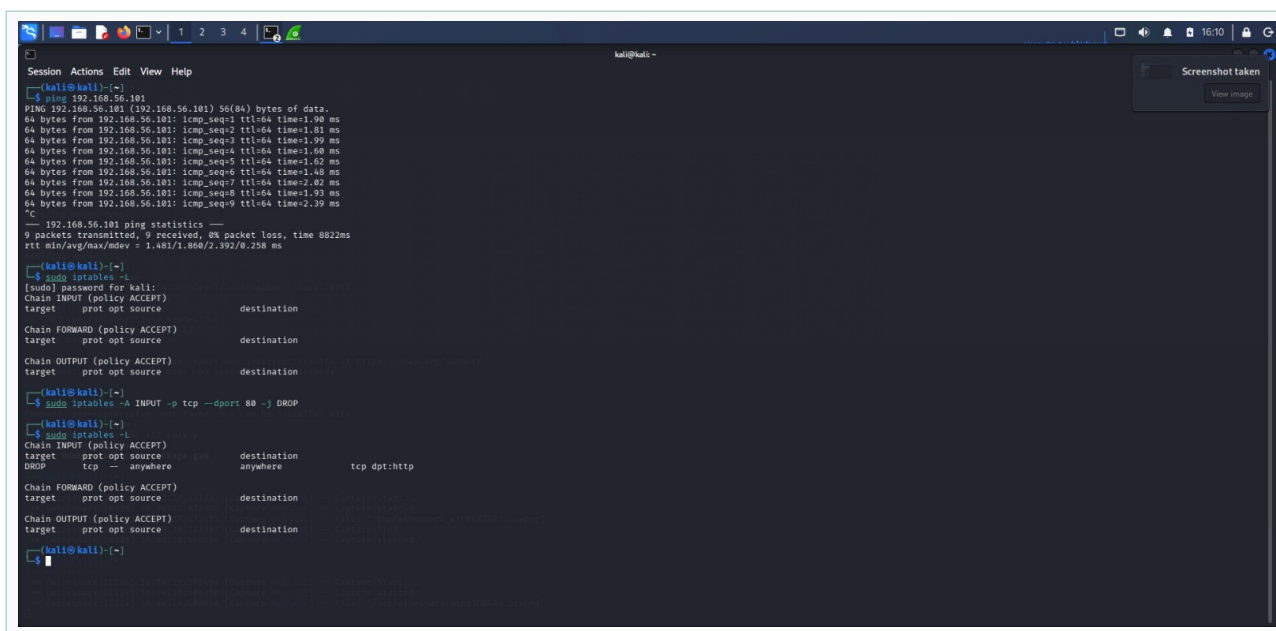


Fig 17 – Final iptables state: firewall rule persisted; INPUT chain shows DROP tcp dpt:http

8. Vulnerability Summary

The following table summarises the critical and high-severity services discovered during the scanning phase. All vulnerabilities are well-known and intentionally present in Metasploitable2 for educational exploitation.

Port/Service	Version	Severity	CVE / Issue	Risk
21 – FTP (vsftpd)	2.3.4	CRITICAL	CVE-2011-2523 Backdoor	Remote shell
22 – SSH	OpenSSH 4.7p1	HIGH	Outdated version	Brute-force risk
23 – Telnet	Linux telnetd	CRITICAL	Plaintext auth	Credential sniff
25 – SMTP	Postfix smtpd	MEDIUM	Open relay possible	Spam abuse
80 – HTTP	Apache 2.2.8	HIGH	CVE-2007-6750+	DoS / WebDAV
139/445 – SMB	Samba 3.x–4.x	CRITICAL	CVE-2017-7494 EternalBlue	RCE
512–514 – r*	netkit-rsh	CRITICAL	Unauthenticated	Remote exec
1524 – bindshell	Metasploitable	CRITICAL	Backdoor root shell	Full takeover
3306 – MySQL	5.0.51a	HIGH	CVE-2009-2446	SQL injection
5432 – PostgreSQL	8.3.0–8.3.7	HIGH	CVE-2009-0922	Priv escalation
5900 – VNC	3.3	HIGH	Weak auth / no TLS	Screen hijack
6667 – IRC	UnrealIRCd	CRITICAL	CVE-2010-2075 Backdoor	Remote shell
8180 – HTTP	Tomcat/Coyote 1.1	MEDIUM	Default creds likely	Admin access

9. Mitigation Recommendations

1. Disable/Remove Backdoored Services

Remove vsftpd 2.3.4 and UnrealIRCd immediately; replace with patched versions. Audit all installed packages against known-malicious version databases.

2. Disable Legacy Plaintext Protocols

Remove Telnet (port 23) and r-services (512–514). Use SSH with key-based authentication only. Disable FTP; use SFTP or FTPS instead.

3. Patch or Remove Samba

Upgrade Samba to a version not affected by EternalBlue (CVE-2017-7494). If SMB is not required, disable it with: `sudo systemctl disable smbd nmbd`.

4. Close the Metasploitable Bind-Shell

Port 1524 exposes a direct root shell with no authentication. This must be removed in any production context. Audit `/etc/inetd.conf` and `/etc/services`.

5. Database Hardening

MySQL and PostgreSQL should not be exposed on all interfaces. Bind to localhost only, enforce strong passwords, and apply all available patches.

6. Deploy iptables / nftables Rules

As demonstrated in Phase 5, iptables can be used to block unnecessary ports. A minimal allow-list policy (DROP by default, only permit required services) is recommended for all production systems.

7. Enable Network Monitoring

Use Wireshark, Zeek, or Suricata for continuous packet analysis. LLMNR and mDNS should be disabled if not needed to reduce attack surface.

8. Apply Principle of Least Privilege

Services should run as non-root users where possible. VNC should require strong authentication and be tunnelled through SSH.

10. Conclusion

This lab exercise provided comprehensive hands-on exposure to the foundational workflow of network-based penetration testing and host security hardening. Beginning with passive host discovery and progressing through active port scanning, service fingerprinting, OS detection, packet-level traffic analysis, and firewall rule management, each phase built upon the previous to construct a complete picture of the target's attack surface.

The Metasploitable2 target was found to expose **23 open TCP ports** running critically vulnerable software across multiple service categories — file transfer, remote access, web services, databases, and inter-process communication. The OS was positively fingerprinted as **Linux kernel 2.6.X** (Ubuntu 8.04). Wireshark confirmed real-time visibility into ICMP, ICMPv6, and IGMPv3 traffic. iptables was successfully used to enforce a firewall rule blocking HTTP (port 80) inbound traffic.

The skills demonstrated — reconnaissance, Nmap scanning, Wireshark packet analysis, and iptables firewall configuration — form the essential toolkit of any cybersecurity professional. This lab reinforces the importance of attack-surface minimisation, service hardening, and continuous network monitoring in real-world environments.

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June 2026

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